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Experiment no. 04

Aim: To set up a Continuous Integration Pipeline using Jenkins/GitHub Actions.

LO Mapped: LO3

Theory

Continuous Integration

Continuous Integration (CI) is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development. Instead of integrating code changes infrequently and manually, CI encourages frequent integration of small changes, each automatically built and verified, so that problems are caught early rather than accumulating.

Jenkins

Jenkins is an open-source automation server that facilitates Continuous Integration and Continuous Delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supports a wide range of plugins and integrations, making it a versatile tool for automating tasks in software development.

What is a Pipeline in Jenkins?

A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a CI/CD environment. In simple terms, it is a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key benefits of a Jenkins Pipeline:

- Automates CI/CD processes
- Provides better visibility into the build process
- Offers version control of build logic (Pipeline as Code)
- Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines

There are two main types of syntax used to define Jenkins pipelines:

1. Declarative Pipeline — a simpler and more structured syntax, designed for users who want an easy-to-read format. It was introduced later to make pipelines more readable and maintainable.

- Uses a pipeline {} block
- Organized into stages and steps
- Easier error handling
- Better validation in the Jenkins UI

2. Scripted Pipeline — a more flexible and powerful syntax, written using Groovy-based code, giving full control over the build process but less beginner-friendly.

- Uses a node {} block
- Everything is written as code
- Allows complex scripting and logic
- Less user-friendly for beginners

PART A: JENKINS INSTALLATION

The following steps were followed to install and set up Jenkins:

#	Step / Command	Description
1	`java -version`	Verified Java (JDK) was already installed on the system (Java 23.0.2), which Jenkins requires to run.
2	`(download jenkins.war)`	The Jenkins WAR file was downloaded from the official Jenkins website and saved to the Desktop.
3	`cd Desktop`	Navigated to the folder containing the downloaded jenkins.war file (saved as "jenkins (1).war").
4	`java -jar "jenkins (1).war"`	Attempted to start Jenkins; since Java 23 is newer than Jenkins' officially supported versions, this returned a warning suggesting the --enable-future-java flag.
5	`java -jar "jenkins (1).war" --enable-future-java`	Re-ran the command with the suggested flag to bypass the Java-version compatibility warning. Jenkins then failed to bind to port 8080 because it was already in use on the system.
6	`java -jar "jenkins (1).war" --enable-future-java --httpPort=8081`	Re-ran the command on an alternate port (8081) to avoid the port conflict. Jenkins started successfully and generated an initial admin password.
7	`http://localhost:8081`	Opened the Jenkins Web UI in a browser, reaching the "Unlock Jenkins" setup wizard.
8	`(enter initial admin password)`	Copied the auto-generated password from the terminal output and entered it to unlock Jenkins.
9	`Install suggested plugins`	Selected the default recommended plugin set during the Customize Jenkins step.
10	`Create First Admin User`	Created a permanent admin account (username, password, full name, email) to replace the temporary initial password.
11	`Confirm Jenkins URL → Dashboard`	Confirmed the instance URL and completed setup, arriving at the Jenkins dashboard.

Note: The system's installed Java version (23.0.2) was newer than Jenkins' officially supported range, requiring the --enable-future-java flag to proceed. Additionally, the default port 8080 was already in use on the system, so Jenkins was started on port 8081 instead using the --httpPort flag. Both issues were resolved without needing to modify the underlying Java installation.

PART B: PIPELINE CREATION

1. Declarative Pipeline

A new Pipeline item named testPipeline was created from the Jenkins dashboard (New Item → Pipeline), and the following Declarative Pipeline script was entered in the Pipeline section:

```
pipeline {
    agent any

    stages {
        stage('Hello') {
```

```
            steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

After saving and running the pipeline using Build Now, the Console Output confirmed successful execution:

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\testPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

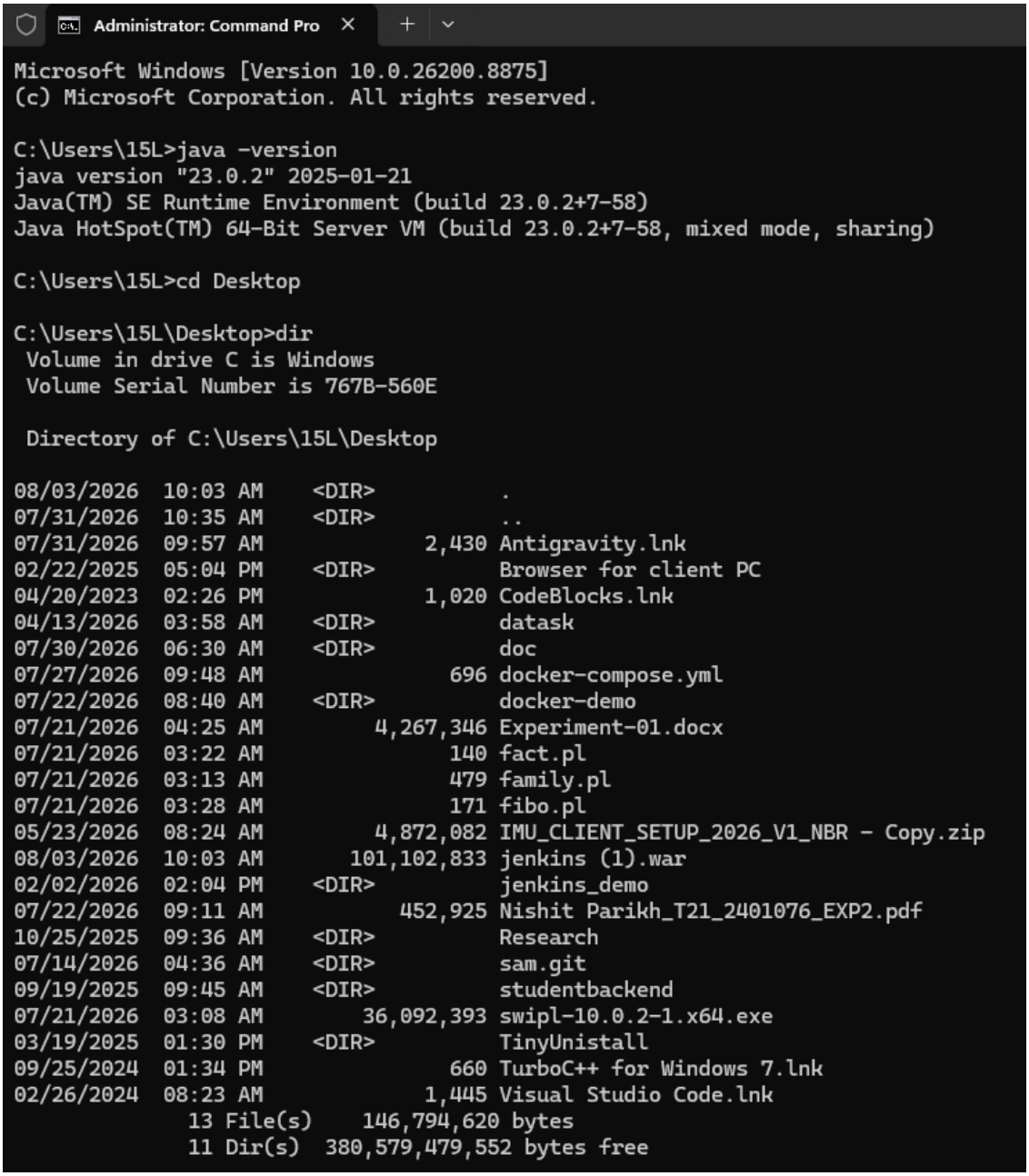
2. Scripted Pipeline

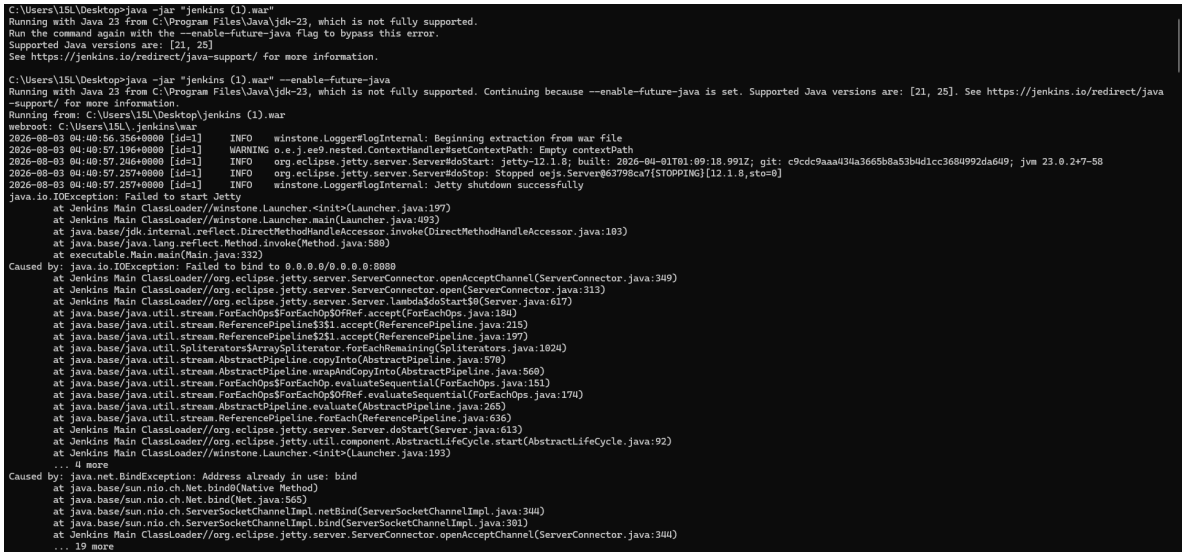
A second Pipeline item named testPipeline2 was created in the same way, this time using Scripted Pipeline syntax:

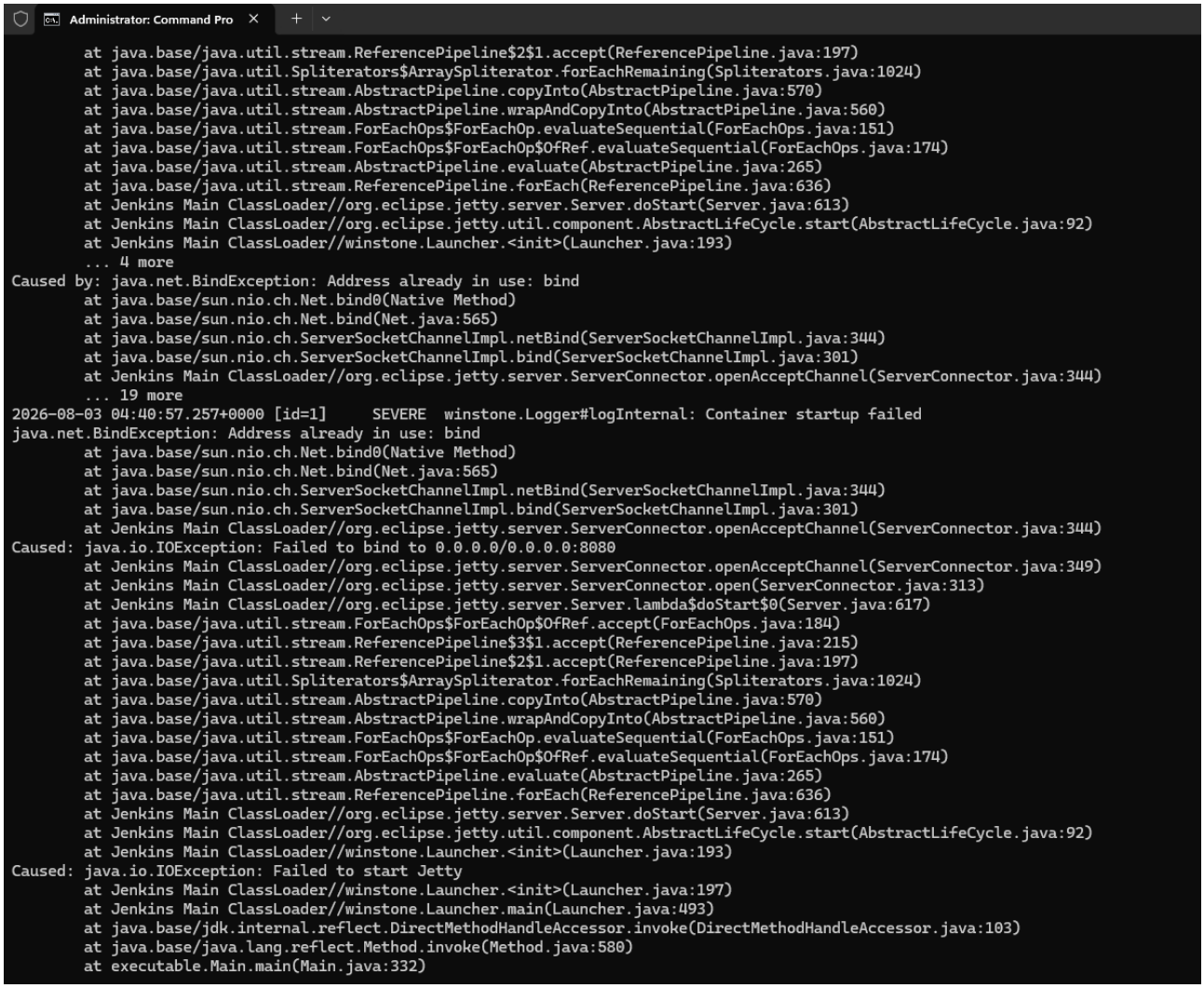
```
node {
    stage('Hello') {
        echo 'Hello from Scripted Pipeline!'
    }
}
```

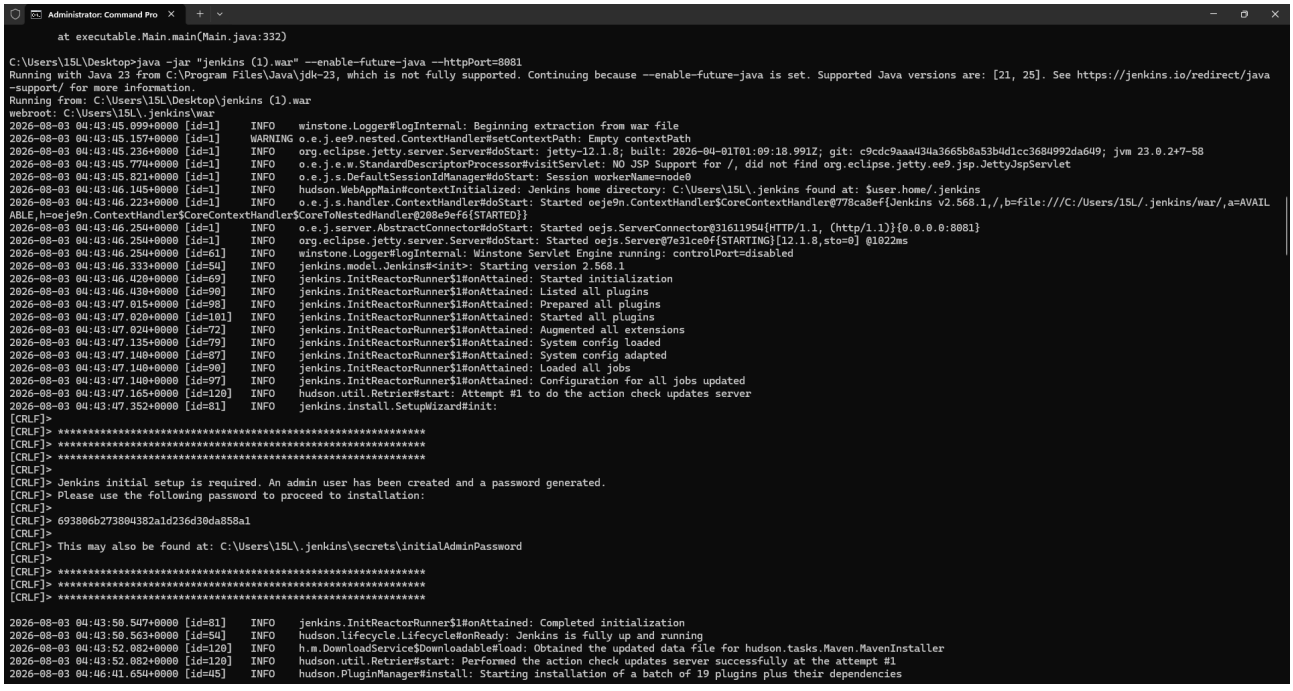
This pipeline was similarly built using Build Now, and its Console Output confirmed successful execution with the message "Hello from Scripted Pipeline!" and a final status of Finished: SUCCESS.

OBSERVATIONS / SCREENSHOTS



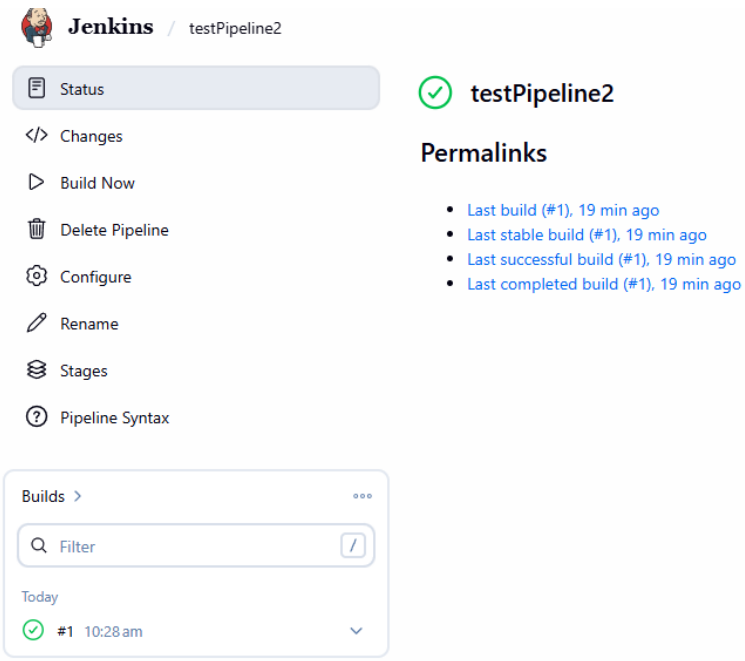



Administrator: Command Pro
at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:197)
at java.base/java.util.Spliterators$ArraySpliterator.forEachRemaining(Spliterators.java:1024)
at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:570)
at java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:560)
at java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
at java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:265)
at java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:636)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.doStart(Server.java:613)
at Jenkins Main ClassLoader//org.eclipse.jetty.util.component.AbstractLifeCycle.start(AbstractLifeCycle.java:92)
at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:193)
... 4 more
Caused by: java.net.BindException: Address already in use: bind
at java.base/sun.nio.ch.Net.bind0(Native Method)
at java.base/sun.nio.ch.Net.bind(Net.java:565)
at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
... 19 more
2026-08-03 04:40:57.257+0000 [id=1] SEVERE winstone.Logger#logInternal: Container startup failed
java.net.BindException: Address already in use: bind
at java.base/sun.nio.ch.Net.bind0(Native Method)
at java.base/sun.nio.ch.Net.bind(Net.java:565)
at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
Caused: java.io.IOException: Failed to bind to 0.0.0.0/0.0.0.0:8080
at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:349)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.open(ServerConnector.java:313)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.lambda$doStart$0(Server.java:617)
at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:184)
at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:215)
at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:197)
at java.base/java.util.Spliterators$ArraySpliterator.forEachRemaining(Spliterators.java:1024)
at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:570)
at java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:560)
at java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
at java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:265)
at java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:636)
at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.doStart(Server.java:613)
at Jenkins Main ClassLoader//org.eclipse.jetty.util.component.AbstractLifeCycle.start(AbstractLifeCycle.java:92)
at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:193)
Caused: java.io.IOException: Failed to start Jetty
at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
at java.base/java.lang.reflect.Method.invoke(Method.java:580)
at executable.Main.main(Main.java:332)


Administrator: Command Pro
at executable.Main.main(Main.java:332)
C:\Users\15L\Desktop>java -jar "jenkins (1).war" --enable-future-java --httpPort=8081
Running with Java 23 from C:\Program Files\Java\jdk-23, which is not fully supported. Continuing because --enable-future-java is set. Supported Java versions are: [21, 25]. See https://jenkins.io/redirect/java-support/ for more information.
Running from: C:\Users\15L\Desktop\jenkins (1).war
webroot: C:\Users\15L\.jenkins\war
2026-08-03 04:43:45.099+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-08-03 04:43:45.157+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-03 04:43:45.236+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 23.0.2+7-58
2026-08-03 04:43:45.774+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-03 04:43:45.821+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-08-03 04:43:46.145+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\15L\.jenkins found at: $user.home/.jenkins
2026-08-03 04:43:46.223+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@778ca8ef{Jenkins v2.568.1,/,b=file:///C:/Users/15L/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@208e9ef6{STARTED}}
2026-08-03 04:43:46.254+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@31611954{HTTP/1.1, (http/1.1)}{0.0.0.0:8081}
2026-08-03 04:43:46.254+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@7e31ce0f{STARTING}[12.1.8,sto=0] @1022ms
2026-08-03 04:43:46.254+0000 [id=61] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-03 04:43:46.333+0000 [id=54] INFO jenkins.model.Jenkins$<init>: Starting version 2.568.1
2026-08-03 04:43:46.420+0000 [id=69] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-03 04:43:46.430+0000 [id=90] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-03 04:43:47.015+0000 [id=98] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-03 04:43:47.020+0000 [id=101] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-03 04:43:47.024+0000 [id=72] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-03 04:43:47.135+0000 [id=79] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-03 04:43:47.140+0000 [id=87] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-03 04:43:47.140+0000 [id=90] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-03 04:43:47.140+0000 [id=97] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-03 04:43:47.165+0000 [id=120] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-03 04:43:47.352+0000 [id=81] INFO jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> **
[CRLF]> **
[CRLF]> **
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 693806b273804382a1d236d30da858a1
[CRLF]>
[CRLF]> This may also be found at: C:\Users\15L\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> **
[CRLF]> **
[CRLF]> **
2026-08-03 04:43:50.547+0000 [id=81] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-03 04:43:50.563+0000 [id=54] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-03 04:43:52.082+0000 [id=120] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-08-03 04:43:52.082+0000 [id=120] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
2026-08-03 04:46:41.654+0000 [id=45] INFO hudson.PluginManager#install: Starting installation of a batch of 19 plugins plus their dependencies


Jenkins / testPipeline
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
Filter
Today
#2 10:23 am
#1 10:23 am
testPipeline
Add description
Permalinks
Last build (#2), 22 min ago
Last stable build (#2), 22 min ago
Last successful build (#2), 22 min ago
Last completed build (#2), 22 min ago


Jenkins / testPipeline / #1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Next Build
Console
Download
Copy
View as plain text
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\testPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


Jenkins / testPipeline2
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
Filter
Today
#1 10:28 am
testPipeline2
Permalinks
Last build (#1), 19 min ago
Last stable build (#1), 19 min ago
Last successful build (#1), 19 min ago
Last completed build (#1), 19 min ago



CONCLUSION: LO3

In this experiment, we learned about Continuous Integration and the process of installing Jenkins, including resolving real-world setup issues such as Java version compatibility and port conflicts. We also learned how to create Jenkins Pipelines to automate the build, test, and deployment process, exploring both Declarative and Scripted pipelines and understanding their structure, benefits, and use cases. By implementing both pipeline types and successfully running them, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.